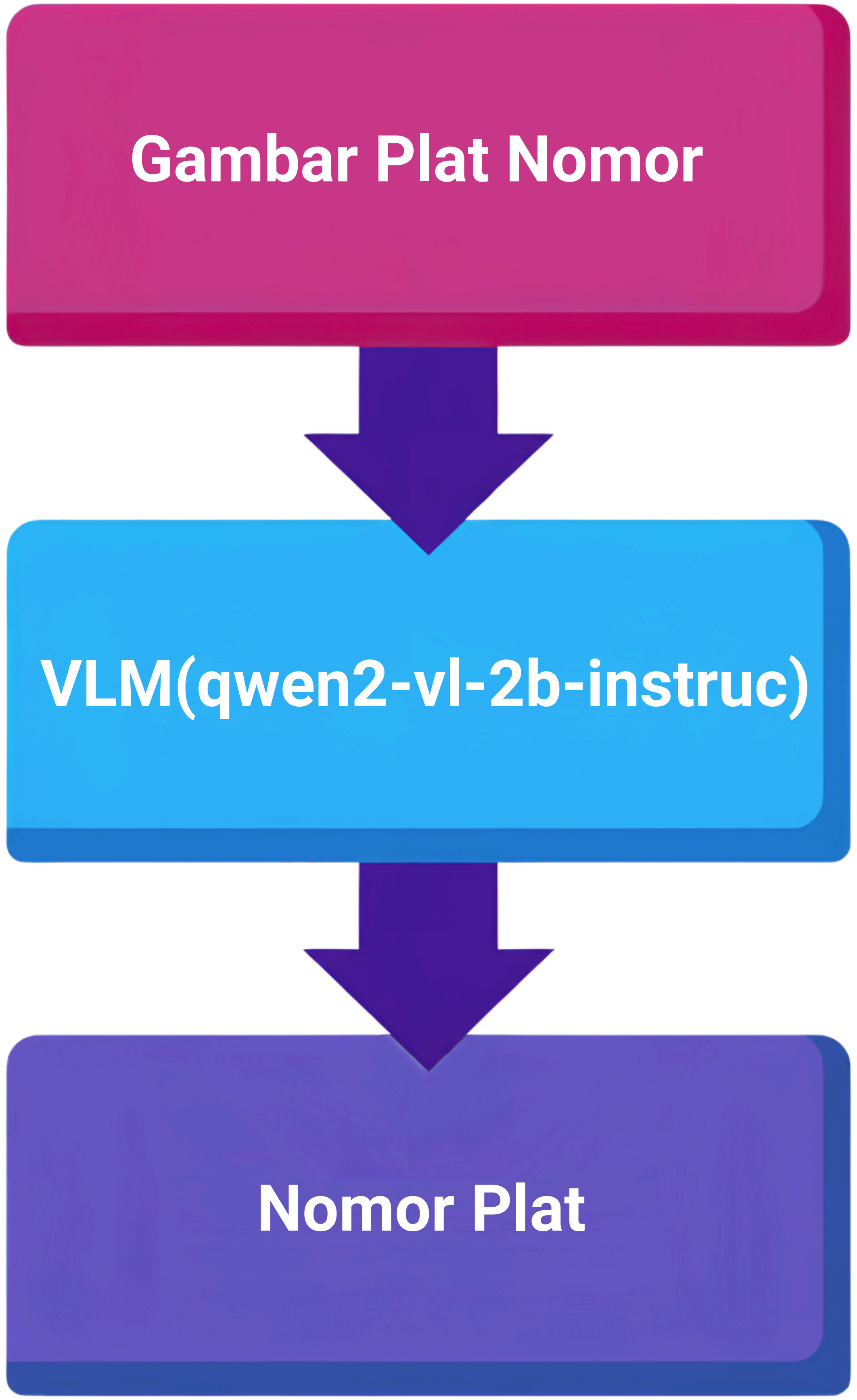
Implementasi OCR Plat Nomor Kendaraan Menggunakan Visual Language Model (VLM) Berbasis LM Studio dan Python




```
"role": "user",
"content": [
    {
        "type": "text",
        "text": (
            "What is the license plate number shown in this image? "
             "Respond only with the plate number."

    },
    {
        "type": "image_url",
        "image_url": {
            "url": f"data:{mime};base64,{image_data}"
```


```python
# Konfigurasi LM Studio
client = OpenAI(
    base_url="http://127.0.0.1:1234/v1",
    api_key="lm-studio",
    timeout=300
)

MODEL_NAME = "qwen2-vl-2b-instruct"
```

```python
with open(image_path, "rb") as f:
    image_data = base64.b64encode(f.read()).decode()
```

```python
response = client.chat.completions.create(
```

```python
# Folder Dataset
IMAGE_FOLDER = "dataset/images/test"
GROUND_TRUTH_FILE = "ground_truth.csv"

# Membaca Ground Truth
gt_df = pd.read_csv(GROUND_TRUTH_FILE)
```

Cara kerja integrasi LMStudio dan Python

Evaluasi Hasil OCR

Rumus Character Error Rate

$$CER = \frac{S + D + I}{N}$$

Keterangan:

- **S (Substitution)** = jumlah karakter yang salah dikenali.
- **D (Deletion)** = jumlah karakter yang hilang.
- **I (Insertion)** = jumlah karakter tambahan.
- **N** = jumlah karakter pada Ground Truth.

```
# CER
cer_score, S, D, I, N = calculate_cer(
    ground_truth,
    prediction
)
```

```
Processing : test001_1.jpg
Ground Truth : B9140BCD
Prediction   : B9140BCD
S = 0
D = 0
I = 0
N = 8
CER = 0.0
```

Proses inferensi dan evaluasi

Contoh Hasil Berhasil

```
Processing : test001_1.jpg
Ground Truth : B9140BCD
Prediction   : B9140BCD
S = 0
D = 0
I = 0
N = 8
CER = 0.0
```

$$CER = \frac{0+0+0}{8} = 0$$

Contoh Hasil Gagal

```
Processing : test001_2.jpg
Ground Truth : B2407UZO
Prediction   : B2407UZ
S = 0
D = 1
I = 0
N = 8
CER = 0.125
```

$$CER = \frac{0+1+0}{8} = 0.125$$

Penjelasan hasil CER dari contoh sukses dan gagal